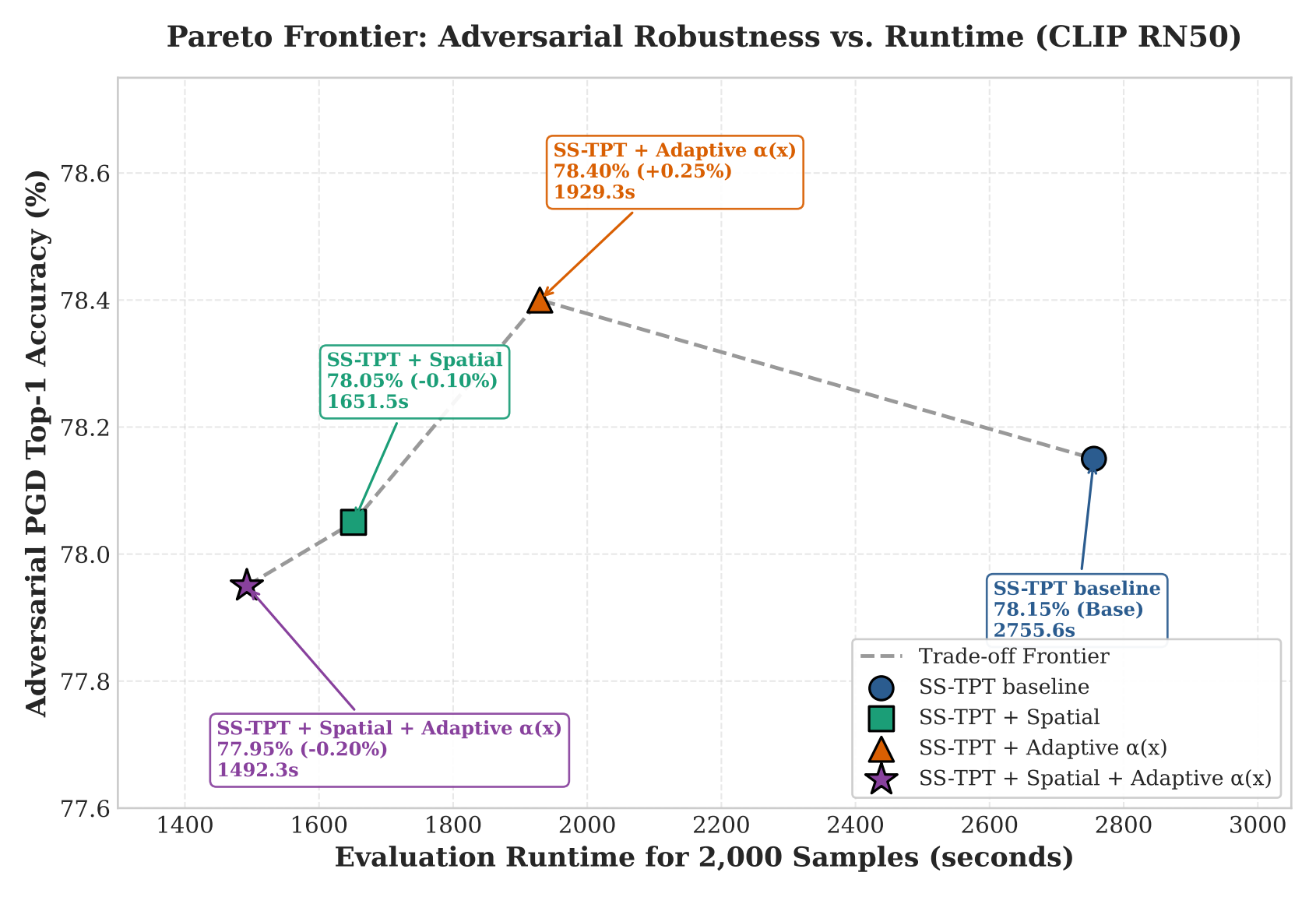
Pareto Frontier: Adversarial Robustness vs. Runtime (CLIP RN50)
Adversarial PGD Top-1 Accuracy (%)
78.6
78.4
78.2
78.0
77.8
77.6
1400
1600
1800
2000
2200
2400
2600
2800
3000
Evaluation Runtime for 2,000 Samples (seconds)
SS-TPT + Adaptive α(x)
78.40% (+0.25%)
1929.3s
SS-TPT + Spatial
78.05% (-0.10%)
1651.5s
SS-TPT baseline
78.15% (Base)
2755.6s
SS-TPT + Spatial + Adaptive α(x)
77.95% (-0.20%)
1492.3s
Trade-off Frontier
SS-TPT baseline
SS-TPT + Spatial
SS-TPT + Adaptive α(x)
SS-TPT + Spatial + Adaptive α(x)